

Mixed Computing Trap Special

fraction + decimal Cross-Topic · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" 5th Volume A Unit 1 + Modern Education 5th Class A Unit 1-3

Core Traps: □ T2 Decimal Operations · T9 Fractional Operations - Non-uniform format direct calculation · Decimal point misalignment

SSPA Association: ● **Extremely High Frequency** Approximately 25% of the scored test paper contains a mixture of fractions + decimals, which is the most likely unit to lose points in the entire paper

Prerequisite knowledge: Chapter 7-10 (four principles of fractions) · Chapter 12-13 (multiplication and division of decimals) · Chapter 16 (comprehensive fractions and decimal equations)

Objectives of this class: ❶ Unified format strategy (convert all fractions vs. all decimals) ❷ Mix the order of four arithmetic operations ❸ Cross-topic application questions

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Convert 0.5 into a fraction (simplest form)	Basic	
2	Convert $\frac{3}{4}$ into decimal	Basic	
3	Calculate $0.25 + \frac{1}{2} = ?$ (Tip: unify the format first)	Advanced	
4	Calculate $\frac{3}{5} \times 0.2 = ?$ (Tip: Multiply in a unified format)	Advanced	
5	Judgment: $\frac{1}{3} + 0.5$ Should the fraction be converted first or the decimal converted first? Why?	Advanced	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Mixed Addition of Fractions and Decimals — The strategy of unifying the format first ● SSPA

- ① **Core Principle** |||SEP|||: Fractions and decimals cannot be added directly! The format must be unified first
Strategy 1 (converting fraction method) |||SEP|||: Convert decimals into fractions → common fraction → calculation → reduction. Applicable to: when decimals can be converted into simple fractions (0.5, 0.25, 0.75, 0.2, 0.125, etc.)
- ② **Strategy 2 (conversion to decimals method)** |||SEP|||: Convert fractions into decimals → Align decimal points → Calculate. Applicable to: When fractions can be converted into finite decimals (the denominator is 2,4,5,8,10,20,25...) The key to choosing a strategy |||SEP|||: See which conversion result is more concise. If the denominator is 3,6,7,9, it can usually be converted to a fraction; if the denominator is 2,4,5,8, it can be converted to a decimal.
- ③ **Strategy 2 (conversion to decimals)**: Convert fractions into decimals → Align decimal points → Calculate. Applicable to: when the fraction can be converted into a finite decimal (the denominator is 2,4,5,8,10,20,25...)
- ④ **The key to choosing a strategy**: See which conversion result is more concise. If the denominator is 3,6,7,9, it can usually be converted to a fraction; if the denominator is 2,4,5,8, it can be converted to a decimal.

□ Example of trap detonation (the most important demonstration in this class)

calculate: $\frac{1}{4} + 0.5 = ?$

✗ **Common mistakes (60% students)**

✓ **Correct solution (conversion to fraction method)**

$0.25 + 0.5 = 0.3$ (the decimal point is misplaced)

Directly correct the decimal places of 0.25 and 0.5, and miscalculate as $0.25 + 0.05 = 0.3$

$$0.5 = \frac{1}{2} \rightarrow \frac{1}{4} + \frac{1}{2} = \frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

 **Tip: "If the format is not consistent, unify it first. The denominator 3/6/7/9 can be converted into a fraction. The denominator 2/4/5/8 can be a decimal. The answer should be the simplest."**

 **The most frequent error: adding fractions and decimals directly without conversion, adding the decimal to the numerator as the answer**

 **The second most frequent error: Decimal point alignment error - tenth place should be aligned with tenth place, and hundredth place should be aligned with hundredth place**

Knowledge Point - Worked Examples (write out unified format → calculateprocess)

#	Question	Difficulty	Working Space
Example 1	$0.75 + \frac{1}{4} = ?$		
Example 2	$\frac{3}{5} + 0.2 = ?$		

Knowledge point 2: Mixed multiplication and division of fractions and decimals ●

SSPA required test

Mixed multiplication and division strategies:

① **Recommended strategy: Convert all fractions-** After converting the fractions to decimals, use fraction multiplication (numerator times numerator, denominator times denominator), and finally reduce

② Division: After converting fractions " $\div$ fraction = $\times$ reciprocal"

If converting to a decimal: Pay attention to the rules for the number of decimal places in decimal multiplication (a digit $\times$ b digit = a+b decimal place)

④ **Key judgment |||SEP|||: If the decimal is a recurring decimal (such as 0.333...), it must be converted to a fraction.:** If the decimal is a recurring decimal (such as 0.333...), it must be converted to a fraction.

Example questions (trap demonstration)

Example 3: $\frac{2}{3} \times 0.5 = ?$

example

Example 4: $0.75 \div \frac{3}{4} = ?$

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
6	$\frac{1}{4} \times 0.8 = ?$		
7	$0.6 \times \frac{5}{6} = ?$		
8	$\frac{3}{8} \div 0.25 = ?$		
9	$0.4 \div \frac{2}{5} = ?$		
10	$\frac{1}{2} \times 0.125 = ?$		

Knowledge point 3: Fractions + decimals in four mixtures (first multiplication and division, then addition and subtraction + unified format) ● SSPA Advanced

Mix four strategies (double challenge):

① **Step 1: Follow the order of operations**—Multiplication and division first, then addition and

subtraction, brackets are calculated first if there are parentheses

② **Step 2: Unify the format of segments**—Convert multiplication and division parts to fractions uniformly, and decide to convert addition and subtraction parts to fractions or decimals

③ **Recommended overall strategy:** It is safest to convert all fractions (to avoid the problem of endless decimal divisions)

④ If there are parentheses and the parentheses contain fractions + decimals: first process the format within the parentheses to be unified

example

Example 5: $0.5 + \frac{1}{3} \times 0.6 = ?$

example

Example 6: $(\frac{1}{2} + 0.25) \times \frac{4}{5} = ?$

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	$0.3 + \frac{1}{2} \times 0.4 = ?$		
12	$\frac{2}{3} \times 0.75 + 0.25 = ?$		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, all must do - per question mixed fraction and decimal)

#	Question	Difficulty	Working Space
13	$0.5 + \frac{1}{3} = ?$		
14	$\frac{1}{2} + 0.25 = ?$		
15	$0.75 + \frac{1}{8} = ?$		

16	$\frac{3}{10} + 0.4 = ?$		
17	$0.2 + \frac{2}{5} = ?$		

Advanced layer (total 7 questions,   choose do - mix multiplydivide + four)

#	Question	Difficulty	Working Space
18	$\frac{1}{4} \times 0.6 = ?$		
19	$0.8 \times \frac{3}{4} = ?$		
20	$\frac{5}{6} \div 0.5 = ?$		
21	$0.3 + \frac{1}{4} \times 0.8 = ?$		
22	$\frac{1}{2} \times 0.6 + 0.25 = ?$		
23	$1.25 - \frac{1}{4} \div 0.5 = ?$		
24	$(0.5 + \frac{1}{3}) \times 0.6 = ?$		

 challenge layer (total 5 questions,  choose do—more step mixing + trap)

#	Question	Difficulty	Working Space
25	$0.75 \times \frac{2}{3} + \frac{1}{6} = ?$		
26	$\frac{3}{4} \div 0.25 + 0.5 \times \frac{1}{3} = ?$		
27	$1.2 \times (\frac{5}{6} - 0.25) = ?$		
28	$(0.6 + \frac{1}{5}) \div (\frac{1}{2} - 0.1) = ?$		
29	$0.125 + \frac{3}{8} \times 0.4 - \frac{1}{10} = ?$		

Knowledge point 4: Cross-topic application questions (shopping discounts + recipe portions + engineering measurements) ● SSPA compulsory exam

Four steps to solve the problem: ① Circle keywords ("Total" "After discount" "Each portion")

② Column (Uniform format for fractions and decimals) ③ Calculation (Pay attention to unit conversion)

④ Write a complete answer (Step points will be deducted if you don't write an answer!)

example

Example 7: A cake originally priced at \$80 is now discounted by 20% (i.e. $\frac{4}{5}$). Xiao Ming has 0.6 pieces of cake. How much does he need to pay?

applicationquestionpractice (all must do, column → unified format → calculate → answer sentence)

#	Question	Difficulty	Working Space
30	A pencil $\frac{1}{2}$ yuan, Xiao Ming bought 0.6 pieces (billed on a pro-rata basis). How much does he need to pay? (Answer in decimals)		
31	Recipe requires $\frac{3}{4}$ kg flour and 0.25 kg sugar. How many kilograms do the two materials weigh in total?		
32	A ribbon is 2.5 meters long and uses $\frac{1}{4}$ to make a bow. How many meters of ribbon were used for the bow?		

advancedapplicationquestion (👤🚀 choose do, SSPA Paper 2 common question type)

#	Question	Difficulty	Working Space
33	A jacket, originally priced at \$360, is 25% off (0.75). In addition, use membership $\frac{1}{10}$ coupons to get discounts. How much does Xiao Ming have to pay in the end?		
34	Recipe: Cake requires $\frac{2}{3}$ cups of milk. How many cups of milk are needed if only 0.75 servings (i.e. $\frac{3}{4}$ servings) are made?		
35	The engineering team completed the project in one day SEP . After 0.5 days, another 0.125 of the project was completed. What fraction of the project has been completed? $\frac{1}{8}$. After 0.5 days, another 0.125 of the		

	project was completed. What fraction of the project has been completed?		
36	The water bottle originally contained 0.75 liters of water. After drinking $\frac{1}{3}$ liters, an additional 0.2 liters was added. How many liters does the water bottle have now?		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	$(0.75 + \frac{1}{6}) \div (\frac{2}{3} - 0.25) \times 1.2 = ?$		
 2	A water tank has a capacity of 5 liters. The first injection of $\frac{1}{4}$ is 0.8 of the cylinder and the second injection is 1.25 liters. How many liters are there in the cylinder now?		
 3	Three stores: store A' s discount is SEP , store B' s discount is 0.65, and store C' s discount is SEP . Which one is the cheapest? Please prove it with calculations. (Tip: Convert all to decimals for comparison) $\frac{3}{5}$, the discount for store B is 0.65, and the discount for store C is $\frac{2}{3}$. Which one is the cheapest? Please prove it with calculations. (Tip: Convert all to decimals for comparison)		

V. Class afterhomework

Basic must doquestion (total 6 questions, perquestionmust mix fractionanddecimal)

#	Question	Difficulty	Working Space
H1	$0.5 + \frac{1}{4} = ?$		
H2	$\frac{2}{5} + 0.3 = ?$		
H3	$\frac{3}{4} \times 0.4 = ?$		
H4	$0.6 \div \frac{3}{5} = ?$		

H5	$0.3 + \frac{1}{2} \times 0.4 = ?$		
H6	A kettle holds 1.5 liters of water and uses SEP . How many liters of water were used? $\frac{1}{3}$. How many liters of water were used?		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H7	$(0.5 + \frac{1}{6}) \times \frac{3}{4} = ?$		
H8	Compare $\frac{2}{3} \times 0.75$ and $0.6 \times SEP $, which is larger? How much difference? $\frac{5}{6}$ Which one is larger? How much difference?		
H9	An item costs \$240, first add $\frac{1}{4}$ (i.e. $\times 1.25$), and then get a 15% discount (0.85). What is the final selling price?		

VI. The Lessoncorecommon errorssummary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Fractions and decimals are added directly without conversion : $0.5 + \frac{1}{3} = 1.5/3$ (random combination)	First unify the format: convert all fractions or all decimals, and then calculate
2	Decimal point alignment error : $0.25 + 0.5 = 0.3$	Tenth place to tenth place, percentile to percentile, add 0 if necessary
3	When converting a fraction into a decimal, the divisions cannot be divided but the approximation is taken	The denominator contains 3, 6, 7, 9, etc. → Convert to fraction, do not convert to decimal
4	Mixing four ignores "multiplication and division first, then addition and subtraction"	Circle the multiplication and division parts first, and then add and subtract after completing the calculation.
5	Wrong number of decimal places in decimal multiplication	a place
6	Fraction division without reciprocal conversion	$\div$ Fraction = $\times$ The reciprocal of the fraction (swapping the numerator and denominator)

7	The answer to the application question does not include the unit or the answer is incomplete.	Must write "Answer:..." + unit, otherwise step points will be deducted
---	--	--